

UJJAIN MAHAKUMBH HACKATHON

A National Tech Challenge to build Smarter & Safer Solutions for **Simhastha 2028**

Registration Number : TH1256

Team Name : Rishi Path

Project Name : IMARS – Integrated Multi-Agency Response & Safety System

Problem Statement : Ensure safety of pilgrims through real-time monitoring, preparedness, and emergency response systems

Team Leader : Subhi Tiwari Role : Database Management

Team Member : Ankit Prajapati Role: Backend Development

Team Member : Shreya Lathi Role : Frontend development

Team Member : Krishnapal Mewada Role: Research

Problem Statement

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During events like **Simhastha Kumbh Mela**, over **10 million attendees** visit the congested area. However, challenges arise, including :

🧭 **Navigation Confusion** → Poor signage, unfamiliar terrain, multiple languages cause delays & lost attendees.

🚻 **Sanitation Issues** → Toilets overcrowded, no real-time cleaning alerts.

🚗 **Parking Chaos** → Attendees waste time searching for available space.

🏨 **Hotel Room Uncertainty** → Difficult to know real-time availability.

🏥 **Emergency Gaps** → Lack of quick access to nearby hospitals & beds.

Proposed Solution

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A unified smart system for navigation, real-time reporting, parking ,cleaning etc facilities to manage large gatherings like **Simhastha Kumbh Mela** safely and efficiently.

QR-Based Dual-Facing Website :

🌐 **Seamless Access** → Scan QR → Choose Language → Enter home page.

👥 **Dual Facing Design** → User & Admin separate views.

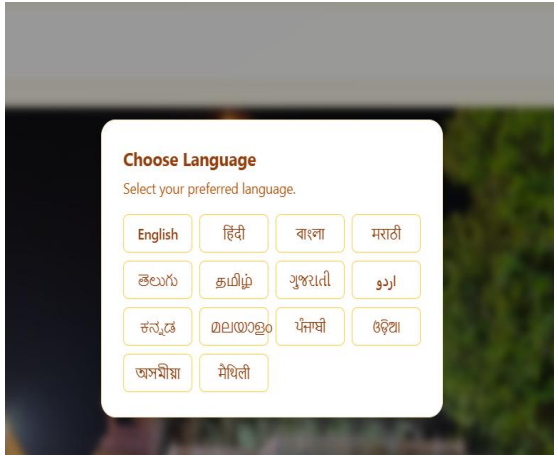
🗺️ **For Users** → Crowd Map, Smart Reporting, Parking, Hotels, Emergency, AI Chatbot.

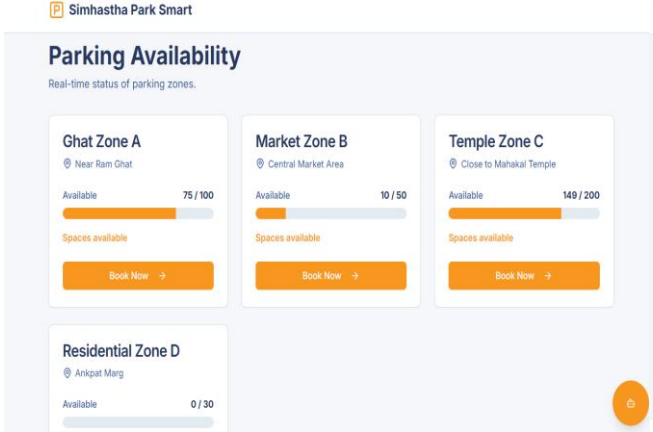
🖥️ **For Admins** → Map Alerts, Sanitation Reports, Task Assignment.

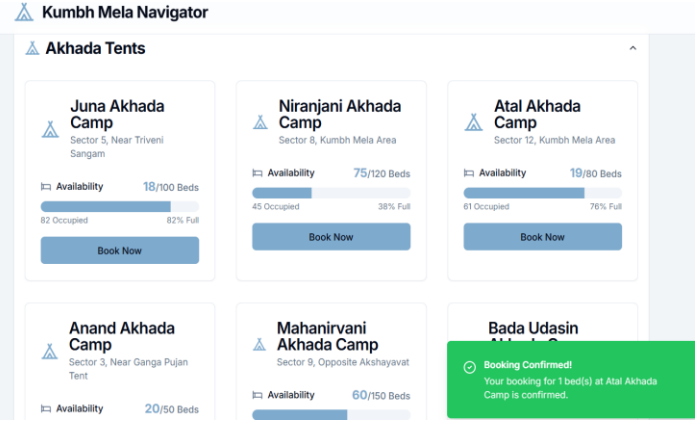
🔄 **Integration** → Combines navigation, facilities, and emergency response in a single platform.

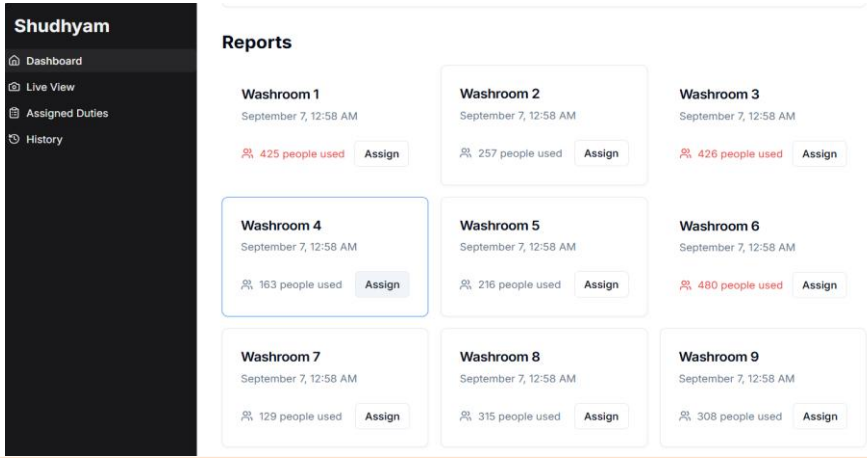
Prototype Demo Screenshots

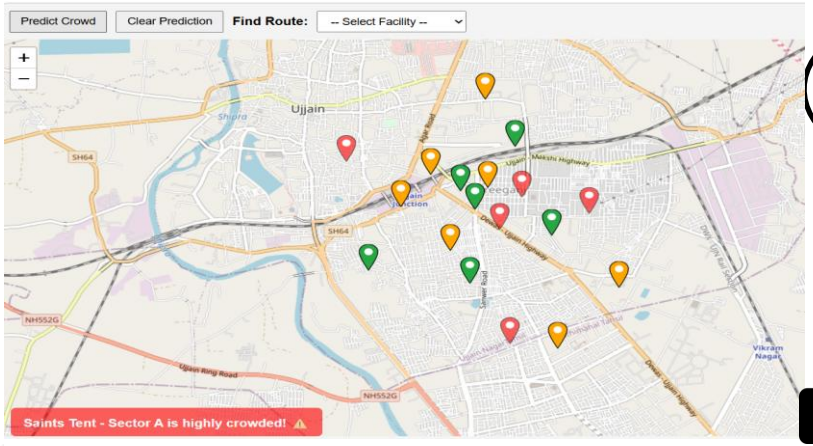
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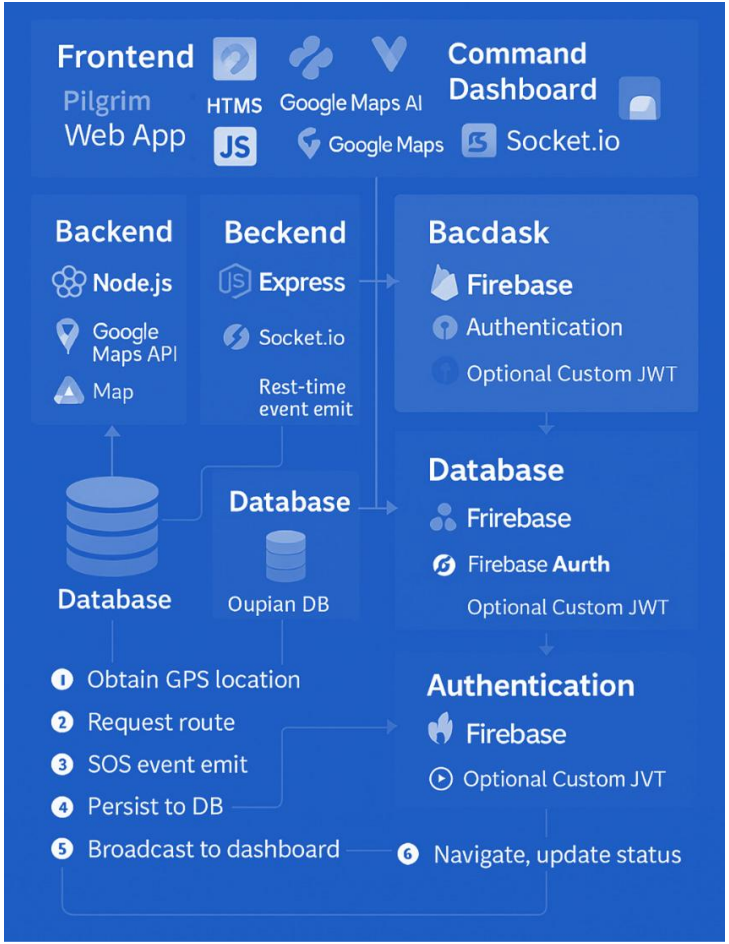




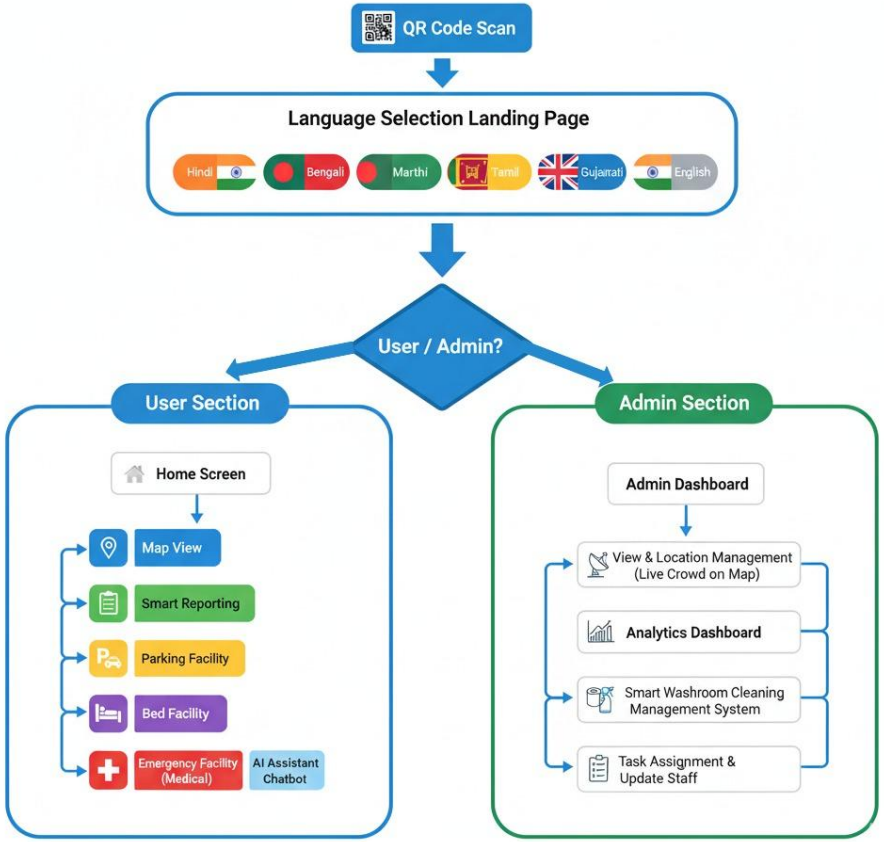
Scan QR and visit website

SCAN ME

Tech Stack



Flow Chart



Impact of System for Large Gatherings

Event Management System for Large Gatherings

Simhastha Kumbh Mela
- Scalable Backend
- GPS-Based Control
- Map-Based Guidance

Multi-Agency Dashboard
- Customized Dashboards
- Real-Time Intel

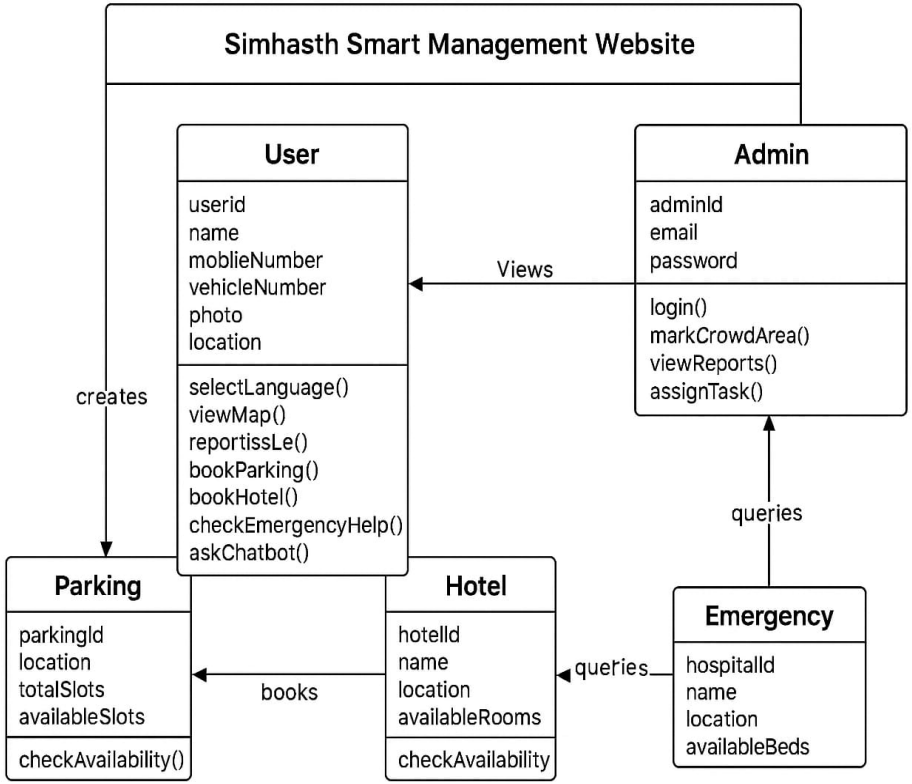
Event Management System

Other Events
- Ganesh Visarjan
- Rath Yatra
- Political Rallies

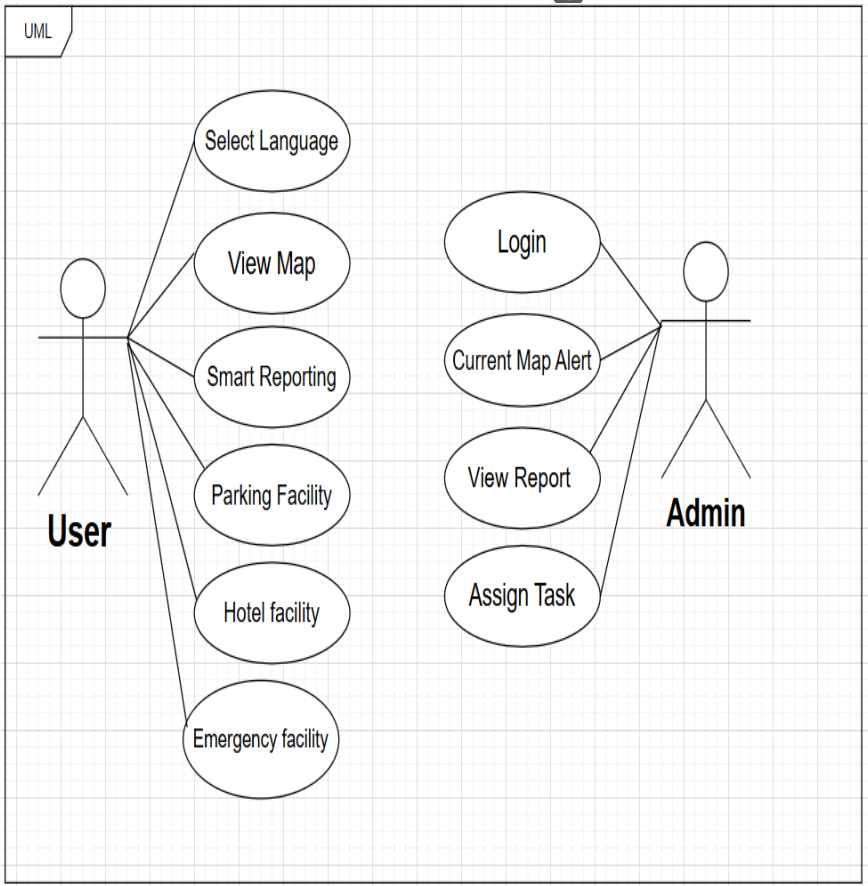
Offline Support Zones
- Static Maps
- Location Tagging



Architecture

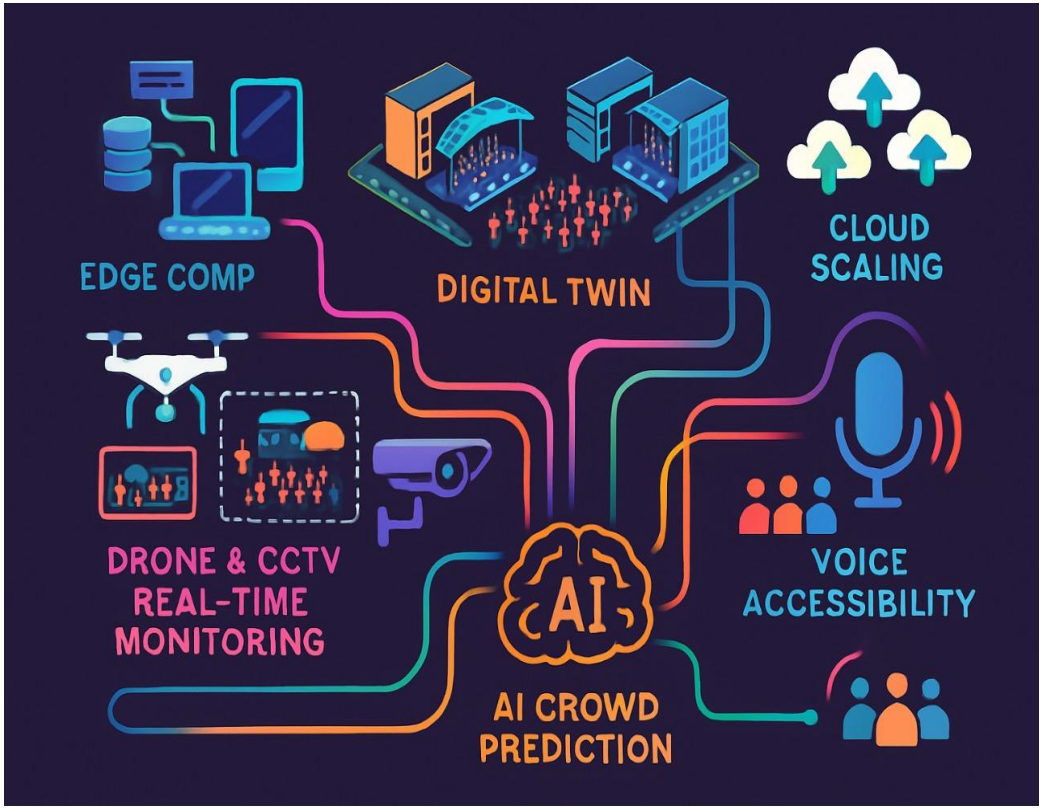


Use Case Diagram



Future Scope

- ⚡ Edge Computing with Ultra-Low Latency
- 🗺️ Digital Twin of Event Area
- ☁️ Cloud Scaling for Millions of Users
- 👴 Voice-Enabled Accessibility
- 🤖 AI-Based Crowd Surge Prediction
- 🛰️ Drone & CCTV Real-Time Monitoring
- 👫 Facial Recognition for Lost Persons



Conclusion

IMARS delivers a transformative advancement in event safety management by integrating real-time monitoring, cloud scalability, and AI-powered predictive analytics to safeguard millions of attendees during large-scale gatherings such as the Simhastha Kumbh Mela. Leveraging edge computing for ultra-low latency, digital twin simulations, and drone + CCTV-based surveillance, the system enables proactive risk mitigation, accelerated emergency response, and data-driven decision-making . Planned enhancements—including voice-enabled accessibility, facial recognition for lost individuals, and AI-driven crowd surge prediction—position IMARS as a scalable, secure, and intelligent safety ecosystem. This evolution ensures operational excellence, regulatory compliance, and resilient public safety infrastructure, setting a global benchmark in smart event management technology.

Department of Science and Technology,
Government of Madhya Pradesh presents



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THANK YOU

Explanation Video Link : click here

https://drive.google.com/file/d/1k1mSrugPCeHn5-Fnfbg_RWtKCqbYvlls/view?usp=drivesdk